$$\Gamma_{\mathrm{therm}}(t) \approx -\pi k_{\mathrm{B}} T t + \ln\left(2\pi k_{\mathrm{B}} T t\right) \approx -\pi k_{\mathrm{B}} T t \gg \Gamma_{\mathrm{fluc}}(t), \tag{5.133}$$

and thus

$$\Gamma(t) \approx -\pi k_{\mathrm{B}} T t. \tag{5.134}$$